Sample Paper 09

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

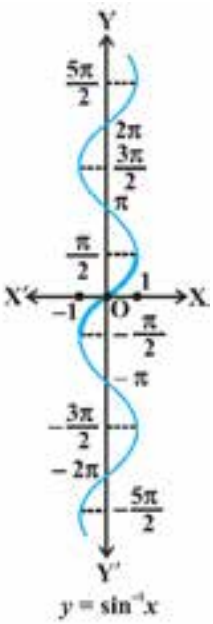
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \sin^{-1}x$?



(a) 0 (b) $\pi/2$

(c) 1 (d) π

2. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b}$ is a vector such that $\vec{a} \cdot \vec{b} = |\vec{b}|^2$ and $|\vec{a} - \vec{b}| = \sqrt{7}$, then $|\vec{b}|$ is equal to

(a) $\sqrt{7}$ (b) $\sqrt{3}$

(c) 7 (d) 3

3. If $f(x) = \log_e(\log_e x)$, then $f'(e)$ is equal to

(a) e^{-1} (b) e

(c) 1 (d) 0

4. The degree of the differential equation $x = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots$, is

(a) 3 (b) 2

(c) 1 (d) not defined

5. The least, value of the function $f(x) = ax + b/x$, $a > 0$, $b > 0$, $x > 0$ is

(a) $\sqrt{ab}$ (b) $2\sqrt{\frac{a}{b}}$

(c) $2\sqrt{\frac{b}{a}}$ (d) $2\sqrt{ab}$

6. If the lines $\frac{1-x}{3} = \frac{y-2}{2\alpha} = \frac{z-3}{2}$ and $\frac{x-1}{3\alpha} = y - 1 = \frac{6-z}{5}$ are perpendicular, then the value of α is

(a) $\frac{-10}{7}$ (b) $\frac{10}{7}$

(c) $\frac{-10}{11}$ (d) $\frac{10}{11}$

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7. A bag A contains 4 green and 3 red balls and bog B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted to be green. The probability that it comes from bag B is

(a) $\frac{2}{7}$

(b) $\frac{2}{3}$

(c) $\frac{3}{7}$

(d) $\frac{1}{3}$

8. Find the area of a curve $xy = 4$, bounded by the lines $x = 1$ and $x = 3$ and X-axis.

(a) $\log 12$

(b) $\log 64$

(c) $\log 81$

(d) $\log 27$

9. If $P(A) = \frac{4}{5}$, and $Q(A \cap B) = \frac{7}{10}$, then $P\left(\frac{B}{A}\right)$ is equal to

(a) $\frac{1}{10}$ (b) $\frac{1}{8}$

(c) $\frac{7}{8}$ (d) $\frac{17}{20}$

10. The point on the curve $x^2 = 2y$ which is nearest to the point $(0,5)$ is

(a) $\left(2\sqrt{2},4\right)$

(b) $\left(2\sqrt{2},0\right)$

(c) $(0,0)$

(d) $(2,2)$

11. The symmetric part of the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 2 \\ 2 & -2 & 7 \end{bmatrix}$ is equal to

(a) $\begin{bmatrix} 0 & -2 & -1 \\ -2 & 0 & -2 \\ -1 & -2 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 4 & 3 \\ 2 & 8 & 0 \\ 3 & 0 & 7 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & 2 \\ -1 & 2 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 4 & 3 \\ 4 & 8 & 0 \\ 3 & 0 & 7 \end{bmatrix}$

12. The value of $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx$ is

(a) $\frac{x^2}{2} + \log|x| - 2x + C$

(b) $\frac{x^2}{2} + \log|x| + 2x + C$

(c) $\frac{x^2}{2} - \log|x| - 2x + C$

(d) None of these

13. The solution of $\frac{dy}{dx} = \frac{ax+g}{by+f}$ represents a circle, when

(a) $a = b$ (b) $a = -b$

(c) $a = -2b$ (d) $a = 2b$

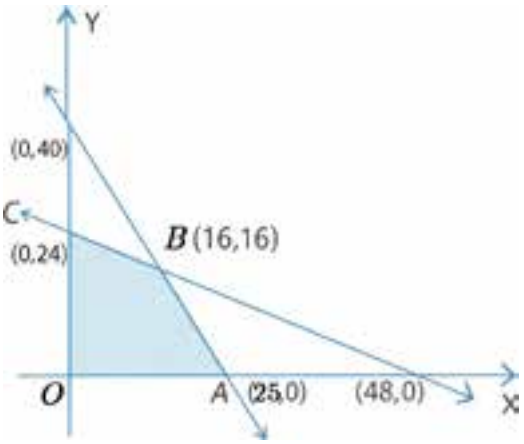
14. If $y = \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ is

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$

(c) 1 (d) -1

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15. In the same figure, if the objective function is given as $z = 4x + y$, then at which vertex of the feasible region does this function attain its maximum value?



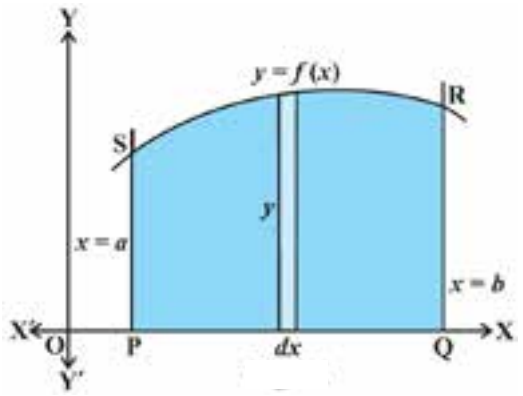
(a) O(0,0)

(b) A(25,0)

(c) B(16,16)

(d) C(0,24)

16. In the figure, the element labeled "dy" represents:



(a) The length of the horizontal strip

(b) The small change in the y-coordinate (thickness of the strip)

(c) The slope of the curve x=g(y)

(d) The total height of the shaded region

17. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units (b) 13 sq units

(c) $13\frac{1}{2}$ sq units (d) 14 sq units

18. The area of the region bounded by the lines $y = mx$, $x = 1$, $x = 2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

(c) 2 (d) 4

19. **Assertion :** $\int \frac{dx}{e^x + e^{-x} + 2} = \frac{1}{e^x + 1} + C$

Reason : $\int \frac{d\{f(x)\}}{\{f(x)\}^2} = -\frac{1}{f(x)} + C$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. **Assertion :** $\int \frac{xe^x}{(x+1)^2} dx = \frac{e^x}{x+1} + C$

Reason : $\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + C$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find distance between (4, 3, 7) and (1, − 1, − 5) ?

22. If $P(A) = \frac{2}{5}$, $P(B) = \frac{3}{5}$, $P(A \cup B) = \frac{3}{4}$, then find $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$

23. If $\vec{a} = 2\hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{b} = -7\hat{i} + 6\hat{j} + 8\hat{k}$ find $\vec{a} \times \vec{b}$

24. Find $\frac{dy}{dx}$ if $y = \tan(x + y)$

OR

Find $\frac{dy}{dx}$ if $y = \sin^3 x \cos^5 x$

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25. Solve – $\frac{dy}{dx} + y\cot x = 2\cos x$

OR

Solve – $xdy + ydx = 0$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. Prove that – $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}\ \vec{b}|^2$

27. Find the rate of change of the area of a circle with respect to its radius r when (i) $r = 3$ cm and (ii) $r = 5$ cm.

28. If $y = \sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$, find $\frac{dy}{dx}$.

29. Examine whether the function $f : R \to R$ is one-one (injective) if $f(x) = x^3$, $x \in R$.

30. Prove that $\tan^{-1}\sqrt{x} - \frac{1}{z}\cos^{-1}\frac{1-x}{1+x}$

OR

Prove that $\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3) = 15$

31. If $A = \begin{bmatrix} 5 & 4 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 & 1 \\ 6 & 8 & 4 \end{bmatrix}$, find AB.

OR

Find the value of x and y if-

$$2\begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve graphically the following L.P. problem:

Maximise $Z = x + y$

Subject to the linear constraints

$$y - 2x - 1 \leq 0$$

$$x \leq 2$$

$$x + y \leq 3$$

and $x, y \geq 0$

33. Solve $(x^2 - y^2)\frac{dy}{dx} = 2xy$

OR

Solve $x\cos x\frac{dy}{dx} + y(x\sin x + \cos x) = 1$

34. Find the probability distribution of

(i) Number of heads in two tosses of a coin.

(ii) Number of tails in the simultaneous tosses of three coins.

(iii) Number of heads in four tosses of a coin.

35. Evaluate $\int \frac{5x-2}{1+2x+3x^2}dx$

OR

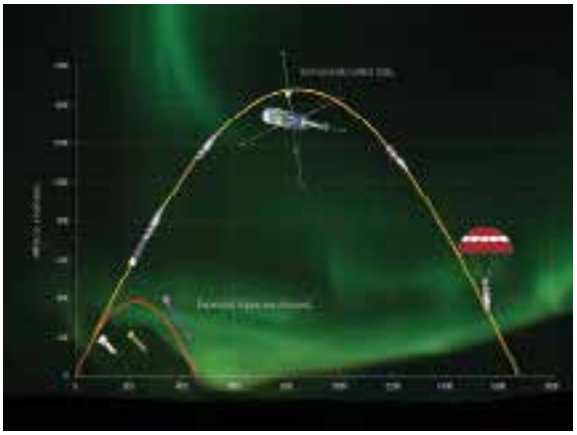
Evaluate: $\int_0^{\frac{\pi}{4}} (\tan x - x)\tan^2 x dx$.

SECTION - E

Case study based questions are compulsory.

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36. Rocket motion is based on Newton's third law, which states that "for every action there is an equal and opposite reaction". Hot gases are exhausted through a nozzle of the rocket and produce the action force. The reaction force acting in the opposite direction is called the thrust force.



The equation of motion of a rocket are : $x = 2t$, $y = -4t$, $z = 4t$, where the time t is given in seconds, and the distance measured is in kilometres.

Based on the above information, answer the following questions.

(i) What is the path of the rocket? Which of the following points lie on the path of the rocket?

(ii) At what distance will the rocket be from the starting point (0, 0, 0) in seconds?

(iii) At certain instant of time, if the rocket is above sea level, where equation of surface of sea is given by $3x - y + 4z = 2$ and position of rocket at that instant of time is $(1, -2, 2)$, then find the image of position of rocket in the sea.

37. The D.A.V. College Managing Committee, familiarly known as DAVCMC, is a non-governmental educational organisation in India and overseas with over 900 schools. 75 colleges and a university. It is based on the ideals of Maharishi Dayanand Saraswati. Full Form of DAV is Dayanand Anglo Vedic.



In a certain city there are 50 colleges and 400 schools. Each school and college has 18 peons, 5 clerks and 1 cashier. Each college in addition has 1 section officer and one librarian. The monthly salary of each of them is as follows:

Peon-₹ 3000, Clerk-₹ 5000, Cashier-₹ 6000, Section Officer-₹ 7000 and Librarian-₹ 9000

Using matrix notation, find

(a) total number of posts of each kind in schools and colleges taken together.

(b) the total monthly salary bill of all the schools and colleges taken together.

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38. Minimum Support Price (MSP) is a form of market intervention by the Government of India to insure agricultural producers against any sharp fall in farm prices. The minimum support prices are announced by the Government of India at the beginning of the sowing season for certain crops on the basis of the recommendations of the Commission for Agricultural Costs and Prices (CACP). MSP is price fixed by Government of India to protect the producer - farmers - against excessive fall in price during bumper production years. The minimum support prices are a guarantee price for their produce from the Government. The major objectives are to support the farmers from distress sales and to procure food grains for public distribution. In case the market price for the commodity falls below the announced minimum price due to bumper production and glut in the market, government agencies purchase the entire quantity offered by the farmers at the announced minimum price.



The Government declare that farmers can get `300 per quintal for onions on 1st July and after that, the price will be dropped by `3 per quintal per extra day. Ramawatar has 80 quintal of onions in the field on 1st July and he estimated that crop is increasing at the rate of 1 quintal per day.

Based on the above information, answer the following questions.

(i) If x is the number of days after 1st July then find the price and quantity of onion in terms of x.

(ii) Find the expression for the revenue as a function of x.

(iii) Find the number of days after 1st July, when Ramawatar attain maximum revenue.

OR

(iv) On which day should Ramawatar harvest the onions to maximum his revenue? What is this maximum revenue?

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